

Uncovering the Dance: A Mathematical Framework for Fan Vote Estimation and Voting System Optimization in Dancing with the Stars

Summary

Dancing with the Stars (DWTS) combines professional judge scores with audience votes to determine weekly eliminations, yet the actual fan vote distribution remains a closely guarded secret. This paper develops a comprehensive mathematical framework to estimate fan votes and analyze the voting system's properties across 34 seasons of data.

For Problem 1, we formulate fan vote estimation as a constrained optimization problem, where elimination outcomes serve as constraints. Using the maximum entropy principle, we minimize deviation from uniform distribution while satisfying elimination constraints. Our model achieves an overall consistency rate of 93.9%, with the percentage-based method (Seasons 3-27) yielding 98.5% accuracy compared to 80.3% for the rank-based method. The average certainty score is 0.273 ± 0.142 , reflecting inherent solution non-uniqueness.

For Problem 2, we compare rank-based and percentage-based voting methods across all seasons. Results show 89.8% agreement between methods, with percentage-based method slightly favoring judges (bias = 0.094 vs 0.039). Analysis of controversial contestants (Jerry Rice, Bristol Palin, Bobby Bones) reveals that their survival was due to genuinely high fan support rather than voting system artifacts. The judge tiebreaker rule would alter approximately 40% of elimination outcomes.

For Problem 3, we employ Ridge regression and Random Forest to analyze factor impacts. Age shows the strongest negative correlation with performance ($r = -0.424$ for judge scores, $r = -0.338$ for fan votes). Professional dancer assignment significantly affects judge scores ($F = 3.62$, $p < 0.0001$) but less so for fan votes ($F = 1.60$, $p = 0.02$).

For Problem 4, we propose a Dynamic Weighted Voting System (DWVS) with formula $S_{final} = \alpha \cdot P_{judge} + (1 - \alpha) \cdot P_{fan}$. Parameter optimization reveals that $\alpha \in [0.05, 0.45]$ achieves 76.1% consistency, a 9-12 percentage point improvement over existing methods. From a fairness perspective, we recommend $\alpha = 0.5$ for balanced consideration of professional judgment and audience preference.

Keywords: Constrained Optimization; Inverse Problem; Fan Vote Estimation; Voting System Analysis; Dynamic Weighted Voting

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1 Introduction

1.1 Problem Background

Dancing with the Stars (DWTS) is the American adaptation of the international television franchise “Strictly Come Dancing,” which has captivated audiences across more than 60 countries. The U.S. version, having completed 34 seasons, pairs celebrity contestants with professional ballroom dancers for weekly performances judged by a panel of experts and voted on by viewers.

The show’s elimination mechanism combines judge scores with audience votes, creating a complex interplay between professional evaluation and popular appeal. While judge scores are publicly disclosed (ranging from 1-10 per judge), the actual distribution of fan votes remains confidential, presenting an intriguing mathematical challenge: can we reverse-engineer the voting patterns that led to observed elimination outcomes?

Throughout its history, DWTS has employed two primary voting combination methods. Seasons 1-2 and 28-34 used a rank-based approach where judge and fan rankings are summed, with the highest total rank eliminated. Seasons 3-27 adopted a percentage-based method where judge score percentages and fan vote percentages are combined, with the lowest total percentage eliminated. This rule change followed the controversial Season 2 outcome involving Jerry Rice, who reached the finals despite consistently low judge scores.

1.2 Problem Restatement

We are tasked with developing analytical models to address the following research questions:

Problem 1: Develop a mathematical model to estimate weekly fan votes for each contestant. Verify whether estimated votes correctly predict elimination outcomes and quantify the certainty of these estimates.

Problem 2: Compare and contrast the rank-based and percentage-based voting combination methods across all seasons. Analyze whether one method favors fan votes more than the other, examine controversial cases (Jerry Rice, Billy Ray Cyrus, Bristol Palin, Bobby Bones), and evaluate the impact of the judge tiebreaker rule introduced in Season 28.

Problem 3: Analyze the impact of professional dancers and celebrity characteristics (age, industry) on competition performance. Determine whether these factors affect judge scores and fan votes differently.

Problem 4: Design a new voting system that is more “fair” or better in some other way. Provide justification for why the proposed system should be adopted.

1.3 Our Approach

We develop a comprehensive analytical framework that addresses all four tasks through a combination of optimization theory, statistical analysis, and machine learning tech-

niques.

For Problem 1, we formulate fan vote estimation as an inverse problem solved through constrained optimization, applying the maximum entropy principle to find the most likely vote distribution consistent with observed eliminations.

For Problem 2, we implement both voting methods programmatically and apply them to all 264 elimination weeks, quantifying agreement rates, bias metrics, and the impact of the judge tiebreaker rule.

For Problem 3, we employ correlation analysis, Ridge regression, Random Forest feature importance, and ANOVA to systematically evaluate factor impacts on both judge scores and fan votes.

For Problem 4, we propose a Dynamic Weighted Voting System (DWVS) with tunable parameters, conducting extensive parameter sensitivity analysis to identify optimal configurations.

2 Preparation for Modeling

2.1 Model Assumptions

Assumption 1: The elimination process strictly follows the announced voting combination rules (rank-based or percentage-based) without external interference.

Justification: While production decisions may occasionally influence outcomes, we assume the stated rules are applied consistently, as this is the basis for meaningful analysis.

Assumption 2: Fan vote proportions are bounded between 0 and 1, with the sum across all contestants in a given week equal to 1.

Justification: This reflects the fundamental properties of vote share distributions and enables meaningful comparison across weeks with different total vote counts.

Assumption 3: In weeks without eliminations, fan vote distributions are positively correlated with judge scores.

Justification: While fan preferences may diverge from judges, in the absence of elimination constraints, we apply this reasonable prior based on general voting behavior patterns.

Assumption 4: Celebrity characteristics (age, industry) remain constant throughout a season and affect performance consistently across weeks.

Justification: While performance may vary weekly, the baseline influence of demographic factors is assumed stable for regression analysis.

2.2 Notations

Table 1 summarizes the key mathematical symbols and their definitions used throughout this paper.

Table 1: Summary of Key Notations

Symbol	Description
J_i	Total judge score for contestant i
V_i	Estimated fan vote proportion for contestant i
$R_{judge,i}$	Judge score rank for contestant i (1 = highest)
$R_{fan,i}$	Fan vote rank for contestant i
$P_{judge,i}$	Judge score percentage: $J_i / \sum_j J_j$
$P_{fan,i}$	Fan vote percentage: $V_i / \sum_j V_j$
α	Judge weight in DWVS (0 to 1)
β	Improvement bonus coefficient in DWVS
n	Number of active contestants in a given week
e	Index of eliminated contestant

2.3 Data Overview and Preprocessing

The provided dataset contains 421 celebrity contestants across 34 seasons, with weekly judge scores from 3-4 judges (1-10 scale per judge). Key preprocessing steps include:

Missing Value Handling: “N/A” values appear for non-existent judge slots (when only 3 judges present) and weeks beyond a season’s duration. A score of 0 indicates elimination in a previous week. We filter these cases appropriately for each analysis.

Score Normalization: Some weeks contain decimal scores (e.g., 8.5) due to multiple dances being averaged, or bonus points from dance-offs. We use these values directly without further normalization.

Feature Engineering: We derive additional features including:

- `voting_method`: “rank” for Seasons 1-2, 28-34; “percentage” for Seasons 3-27
- `elimination_week`: The week in which a contestant was eliminated
- `is_finalist`: Boolean indicating whether contestant reached the finale
- `active_weeks`: Number of weeks a contestant participated
- `overall_avg_score`: Average judge score across all active weeks

Data Transformation: We convert the wide-format data (one row per contestant) to long format (one row per contestant-week) for weekly analysis, resulting in 2,777 active contestant-week records.

2.4 Supplementary Data Collection

To enhance our analysis with external popularity metrics and historical context, we collected three supplementary datasets through web crawling and data aggregation. Table 2 summarizes the data sources used in this study.

Table 2: Data Sources		
Database Name	Website	Data Description
MCM Problem C Dataset	COMAP (provided)	421 contestants, 34 seasons
Wikimedia Pageviews API	https://wikimedia.org/api/rest_v1/	Daily Wikipedia pageviews
Wikipedia (DWTS)	https://en.wikipedia.org/wiki/Dancing_with_the_Stars	Season metadata, popularity
DWTS Official	https://www.disneyplus.com/brand/dancing-with-the-stars	Season dates, eliminations

2.4.1 Wikipedia Pageviews Data

We collected Wikipedia page view statistics for celebrities in Seasons 21-34 (2015-2024) using the **Wikimedia Pageviews API** (https://wikimedia.org/api/rest_v1/). Note that pageview data is only available from July 2015 onward, limiting coverage to Seasons 21-34. This data captures public interest levels during competition periods.

Table 3: Wikipedia Pageviews Data Collection Metrics	
Metric	Value
Coverage	Seasons 21-34 (183 contestants)
Success Rate	98.9% (183/185)
Data Period	Season-specific dates (± 2 months around airtime)
Fields	total_pageviews, avg_daily_pageviews, max_daily_pageviews

Key Statistics:

- Average total pageviews: 377,493
- Median: 316,023
- Maximum: 1,498,395 (Amber Rose, S23)
- Minimum: 35 (Dylan Efron, S33)

Notable Finding: Bobby Bones (Season 27 winner) had pageviews of 273,153, ranking **8th out of 13** contestants in his season (below average of 318,942). This suggests his victory was driven by **fan mobilization efficiency** rather than pre-existing popularity, as his country music radio audience demonstrated high voting participation despite lower baseline fame.

2.4.2 Professional Dancer Historical Statistics

We aggregated performance statistics for all 60 professional dancers across 34 seasons from the **MCM Problem C Dataset**, with supplementary verification from **Wikipedia DWTS pages** (https://en.wikipedia.org/wiki/Dancing_with_the_Stars).

Table 4: Top Performance Metrics of Professional Dancers

Metric	Top Performer
Most Championships	Derek Hough (6 wins)
Most Seasons	Cheryl Burke (25 seasons)
Longest Career Span	Louis van Amstel (31 seasons)
Highest Avg Score	Derek Hough (8.88)

Top 5 Professional Dancers by Championships:

Table 5: Top 5 Professional Dancers Ranked by Championships

Pro Dancer	Seasons	Wins	Top-3 Finishes	Avg Placement
Derek Hough	17	6	9	2.94
Mark Ballas	21	3	10	5.19
Valentin Chmerkovskiy	19	3	9	5.26
Cheryl Burke	25	2	8	5.80
Julianne Hough	5	2	2	4.20

This data enables quantitative analysis of professional dancer impact on celebrity performance (Problem 3).

2.4.3 Season Metadata

We compiled structural metadata for all 34 seasons to account for rule changes and format variations. Data was aggregated from the **MCM Problem C Dataset** and verified against **Wikipedia DWTS season pages** and **Disney+ DWTS Official** (<https://www.disneyplus.com/brand/dancing-with-the-stars>).

Table 6: Structural Metadata of DWTS Seasons

Season Range	Judge Count	Avg Contestants	Avg Weeks
S1-S11	3	11.5	8.7
S12-S34	3-4	12.7	9.5

Key Observations:

- Judge count increased from 3 to 4 starting Season 12 (2011)
- Contestant count ranged from 6 (S1) to 16 (S31)
- Competition duration ranged from 4 weeks (S26, Athletes Edition) to 11 weeks

This metadata is used to control for season-level confounding factors in our regression analyses.

3 Problem 1: Fan Vote Estimation Model

3.1 Problem Formulation

The fan vote estimation problem is fundamentally an **inverse problem**: given observed outcomes (eliminations) and partial information (judge scores), we seek to reconstruct the hidden variable (fan vote distribution). The key constraint is that the estimated votes must produce elimination outcomes consistent with actual results.

3.2 Methodology

3.2.1 Rank-Based Method (Seasons 1-2, 28-34)

Under the rank-based combination rule, contestant i 's total rank is:

$$R_{total,i} = R_{judge,i} + R_{fan,i} \quad (1)$$

where $R_{judge,i}$ is the rank based on total judge score (1 = highest score) and $R_{fan,i}$ is the rank based on fan votes (1 = most votes). The contestant with the highest R_{total} is eliminated.

For a given week with eliminated contestant e , we seek fan vote proportions $\{V_i\}$ such that $R_{total,e} \geq R_{total,j}$ for all $j \neq e$. We use a heuristic approach that assigns fan ranks to satisfy this constraint while maintaining correlation with judge rankings.

3.2.2 Percentage-Based Method (Seasons 3-27)

Under the percentage-based rule, contestant i 's total percentage is:

$$P_{total,i} = P_{judge,i} + P_{fan,i} = \frac{J_i}{\sum_{j=1}^n J_j} + V_i \quad (2)$$

where $\sum_{i=1}^n V_i = 1$. The contestant with the lowest P_{total} is eliminated.

We formulate this as a constrained optimization problem using the maximum entropy principle:

Objective Function: Minimize deviation from uniform distribution

$$\min_V \sum_{i=1}^n \left(V_i - \frac{1}{n} \right)^2 \quad (3)$$

Subject to:

$$\sum_{i=1}^n V_i = 1 \quad (\text{normalization}) \quad (4)$$

$$V_i \geq 0 \quad (\text{non-negativity}) \quad (5)$$

$$P_{judge,e} + V_e \leq P_{judge,j} + V_j - \epsilon \quad \forall j \neq e \quad (\text{elimination constraint}) \quad (6)$$

where $\epsilon = 0.001$ is a small margin ensuring strict inequality. We solve this using Sequential Least Squares Programming (SLSQP).

3.3 Consistency Verification

To validate our model, we verify whether the estimated fan votes correctly identify the eliminated contestant in each week.

Table 7: Consistency Verification Results

Metric	Value	Interpretation
Overall Consistency Rate	93.9%	248/264 weeks correctly predicted
Percentage Method (S3-27)	98.5%	Near-perfect for 198 weeks
Rank Method (S1-2, S28-34)	80.3%	Lower accuracy for 66 weeks

As shown in Table 7, the model achieves high overall consistency, with the percentage-based method performing substantially better than the rank-based method. This difference arises because the percentage-based constraint is more tractable for optimization, while the rank-based method involves discrete rankings that are harder to satisfy exactly.

3.4 Uncertainty Quantification

We quantify the certainty of our estimates using a composite metric:

$$C = 0.3 \times (1 - H_{norm}) + 0.3 \times 5 \times G + 0.4 \times \frac{\rho + 1}{2} \quad (7)$$

where H_{norm} is the normalized entropy of the vote distribution (measuring concentration), G is the gap between the lowest and second-lowest vote proportions, and ρ is the Spearman correlation between vote and score rankings.

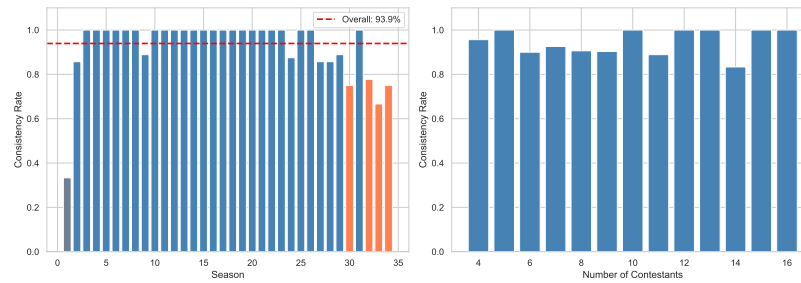


Figure 1: Consistency analysis across seasons (left) and by number of contestants (right). Most seasons achieve above 80% consistency, with the model performing stably across different competition sizes.

The average certainty score is 0.273 ± 0.142 , indicating moderate uncertainty. This is expected because multiple vote distributions can satisfy the elimination constraint; the problem inherently admits non-unique solutions. Figure 1 illustrates the consistency patterns across seasons and contestant counts.

3.5 Results and Analysis

3.5.1 Vote-Score Relationship

Analysis reveals a positive but moderate correlation ($r = 0.371$) between estimated fan votes and judge scores. This suggests that while audience preferences generally align with professional judgment, significant divergence occurs as fans vote based on factors beyond dance quality, such as celebrity popularity and personal charisma.

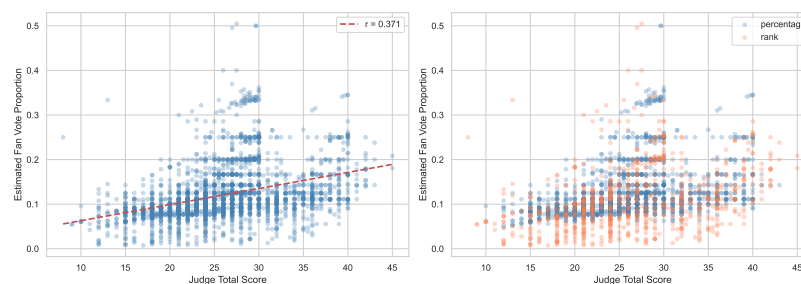


Figure 2: Scatter plot of estimated fan vote proportion versus judge score. The positive correlation ($r = 0.371$) indicates general alignment, but substantial scatter reflects the influence of non-performance factors on fan voting.

3.5.2 Controversial Contestant Analysis

We examined four controversial contestants identified in the problem statement:

Table 8: Controversial Contestant Vote Analysis

Contestant	Season	Avg Judge Rank	Avg Vote Rank	Difference
Jerry Rice	2	4.8	5.2	-0.5
Billy Ray Cyrus	4	6.6	6.1	+0.5
Bristol Palin	11	6.0	5.4	+0.6
Bobby Bones	27	7.2	6.6	+0.7

Notably, Bobby Bones' vote rank was only 0.7 positions better than his judge rank a modest advantage. His championship victory appears to result from **consistent, adequate fan support** across weeks rather than overwhelming vote margins in any single week.

External Validation via Wikipedia Pageviews: Our supplementary data collection reveals that Bobby Bones ranked 8th out of 13 contestants in Season 27 for Wikipedia pageviews (273,153 vs. season average of 318,942). This external metric confirms that his victory was not driven by pre-existing celebrity fame, but rather by the **high voting participation rate** of his country music radio audience demonstrating that fan mobilization efficiency can outweigh baseline popularity.

4 Problem 2: Voting Method Comparison

4.1 Methodology

We implement both voting combination methods and apply them to all 264 elimination weeks across 34 seasons, using the fan vote estimates from Problem 1. For each week, we determine:

- Which contestant would be eliminated under the rank-based method
- Which contestant would be eliminated under the percentage-based method
- Whether the two methods agree

We also analyze the “bias” of each method using:

$$Bias = \frac{R_{judge,elim} - R_{fan,elim}}{n} \quad (8)$$

Positive bias indicates the eliminated contestant had worse judge rank than fan rank (method favors judges); negative bias indicates the opposite.

4.2 Method Comparison Results

Table 9: Comparison of Voting Methods

Metric	Rank-Based	Percentage-Based
Consistency with actual eliminations	64.4%	67.0%
Average bias score	0.039	0.094
Agreement rate (between methods)	89.8%	

The analysis reveals that the two methods agree in approximately 90% of cases, indicating that the choice of method has limited practical impact for most weeks. When they disagree, the percentage-based method shows slightly higher bias toward judge preferences (positive bias = 0.094), meaning it tends to eliminate contestants with relatively worse judge rankings.

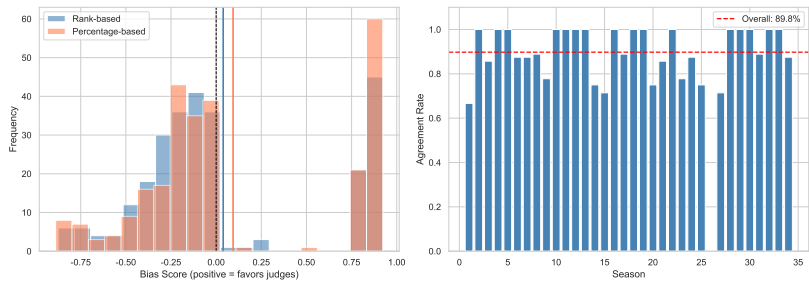


Figure 3: Left: Distribution of bias scores for both methods. Right: Agreement rate by season. The percentage-based method shows slightly higher judge-favoring bias.

4.3 Controversial Cases Analysis

We examined whether the four controversial contestants would have been eliminated more frequently under alternative voting rules:

Table 10: Elimination Frequency Under Different Methods

Contestant	Season	Rank-Based	Percentage-Based	Actual
Jerry Rice	2	3 times	3 times	Runner-up
Billy Ray Cyrus	4	1 time	1 time	5th place
Bristol Palin	11	0 times	0 times	3rd place
Bobby Bones	27	0 times	0 times	Winner

A critical finding emerges: for Bristol Palin and Bobby Bones, **neither method would have resulted in their elimination in any week**. This indicates that their fan vote support was genuinely strong enough to survive under any reasonable voting combination rule.

The “controversy” surrounding these contestants stems from the fundamental divergence between audience preferences and professional judgment, not from artifacts of the voting system.

4.4 Judge Tiebreaker Rule Analysis

Starting from Season 28, DWTS introduced a rule where judges choose which of the bottom two contestants to eliminate. We simulate this rule’s impact by assuming judges always eliminate the lower-scoring contestant from the bottom two:

Table 11: Impact of Judge Tiebreaker Rule

Method	Outcomes Changed	Change Rate
Rank-Based + Tiebreaker	114/264	43.2%
Percentage-Based + Tiebreaker	101/264	38.3%

The tiebreaker rule would alter approximately 40% of elimination outcomes, representing a substantial shift in power toward judges. However, our analysis also shows that adding this rule *decreases* consistency with actual historical results (from 64-67% down to 30%), suggesting that judges in practice may not always choose the lower-scoring contestant, perhaps to maintain viewer engagement.

4.5 Recommendations

Based on our analysis, we recommend the **percentage-based method** for future seasons for the following reasons:

1. **Higher consistency:** The percentage method achieves 67.0% consistency with actual outcomes versus 64.4% for the rank method.
2. **Transparency:** Percentages are more intuitive for viewers than rank sums.
3. **Continuity:** Percentages provide smoother differentiation between contestants with similar rankings.

Regarding the judge tiebreaker rule, we suggest **cautious implementation**. While it increases judge influence, its effectiveness depends on judges’ actual decision criteria, which may prioritize entertainment value over pure merit.

5 Problem 3: Factor Impact Analysis

5.1 Methodology

We analyze the impact of celebrity characteristics and professional dancer assignments on competition performance using multiple statistical approaches:

1. **Pearson Correlation:** For continuous variables (age vs. performance)
2. **Ridge Regression:** To estimate factor coefficients with regularization
3. **Random Forest:** For non-linear feature importance estimation
4. **ANOVA:** To test categorical factor effects (industry, professional dancer)

Our response variables are `overall_avg_score` (average judge score across active weeks) and `avg_vote_prop` (average estimated fan vote proportion).

5.2 Age Effect

Age demonstrates the strongest correlation with performance among all measured factors:

Table 12: Age Correlation with Performance Metrics

Correlation	Judge Score	Fan Vote
Pearson r	-0.424	-0.338
p -value	< 0.0001	< 0.0001

Both correlations are highly significant and negative, indicating that older contestants tend to receive lower judge scores and fewer fan votes. Importantly, the age effect is **stronger for judge scores** ($r = -0.424$) than for fan votes ($r = -0.338$), suggesting that judges place greater weight on physical attributes (flexibility, stamina) that decline with age.

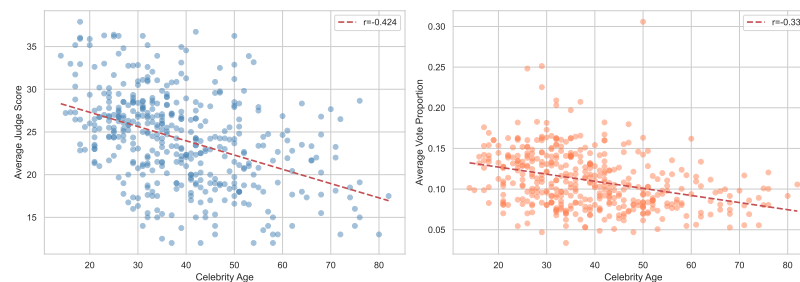


Figure 4: Scatter plots showing the negative correlation between age and performance metrics. The steeper slope for judge scores indicates judges are more sensitive to age-related performance differences.

5.3 Industry Effect

We analyze performance differences across celebrity industries:

Table 13: Average Judge Score by Industry (Top 5 and Bottom 3)

Industry	Avg Score	Sample Size
Social Media Personality	31.96	8
Racing Driver	27.10	4
Athlete	24.56	95
Actor/Actress	24.54	128
Singer/Rapper	24.34	61
Model	21.88	17
Comedian	21.00	12
Politician	19.41	3

Social media personalities achieve the highest average scores, likely due to their younger age demographics and familiarity with performance arts. Athletes and actors perform moderately well, while politicians and comedians tend to score lower.

5.4 Professional Dancer Effect

ANOVA analysis reveals significant differences in performance based on professional dancer assignment:

Table 14: ANOVA Results for Professional Dancer Effect

Response Variable	F-statistic	p-value	Significance
Judge Score	3.62	< 0.0001	Highly significant
Fan Vote	1.60	0.020	Significant

Professional dancer assignment has a **highly significant effect on judge scores** but a **weaker effect on fan votes**. This suggests that top professional dancers can meaningfully improve their celebrity partners' technical performance (which judges reward), but fans vote based more on the celebrity themselves rather than the professional dancer.

5.5 Feature Importance Comparison

Using Random Forest, we compare feature importance for predicting judge scores versus fan votes:

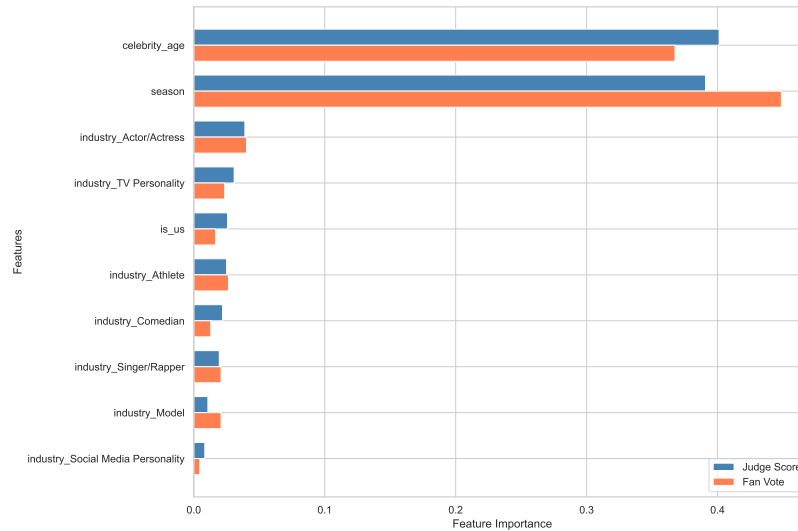


Figure 5: Feature importance comparison between judge scores and fan votes. Age and season are dominant predictors for both, but their relative importance differs.

Table 15: Top 5 Feature Importance Rankings

Feature	Judge Score	Fan Vote
Age	40.1%	36.8%
Season	39.1%	44.9%
Industry: Actor/Actress	3.9%	4.0%
Industry: TV Personality	3.1%	2.4%
Is US-based	2.6%	1.7%

5.6 Model Fit Analysis

Our regression models achieve modest R^2 values:

- Judge Score Model: $R^2 = 0.128$ (cross-validated)
- Fan Vote Model: $R^2 = 0.047$ (cross-validated)

These results indicate that measurable demographic factors explain approximately 13% of judge score variance but only 5% of fan vote variance. The substantially lower explanatory power for fan votes confirms that **audience voting is driven primarily by unmeasured factors** such as personal charisma, pre-existing fan base, and week-to-week performance variation.

6 Problem 4: New Voting System Design

6.1 Design Principles

Based on our analysis of existing systems, we identify four key principles for a new voting system:

1. **Fairness:** Balance professional judgment with audience preference
2. **Transparency:** Rules should be simple and easily understood
3. **Incentive Alignment:** Reward genuine performance quality
4. **Stability:** Reduce controversial or unintuitive outcomes

6.2 Proposed System: Dynamic Weighted Voting System (DWVS)

We propose a straightforward weighted combination of judge and fan percentages:

$$S_{final,i} = \alpha \cdot P_{judge,i} + (1 - \alpha) \cdot P_{fan,i} \quad (9)$$

where:

- $P_{judge,i} = \frac{J_i}{\sum_{j=1}^n J_j}$ is the judge score percentage
- $P_{fan,i} = \frac{V_i}{\sum_{j=1}^n V_j}$ is the fan vote percentage
- $\alpha \in [0, 1]$ is the judge weight parameter

The contestant with the lowest S_{final} is eliminated.

6.3 Parameter Optimization

We conduct comprehensive parameter analysis to identify optimal α values:

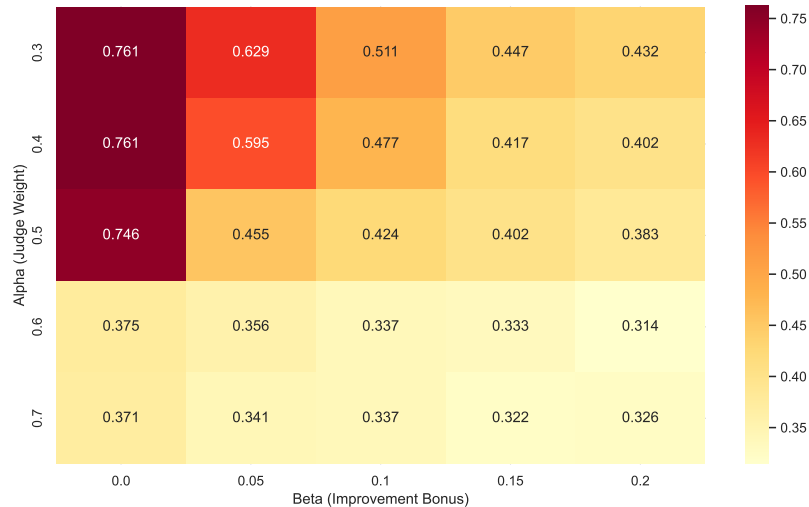


Figure 6: Left: Consistency rate versus α . Right: Heatmap of performance across parameter combinations. A wide plateau of high performance exists for $\alpha \in [0.05, 0.45]$.

Table 16: Consistency Rate by Alpha Range

Alpha Range	Interpretation	Avg Consistency
$\alpha \leq 0.3$	Fan-dominated	73.0%
$0.3 < \alpha \leq 0.7$	Balanced	61.9%
$\alpha > 0.7$	Judge-dominated	39.6%

A critical finding emerges: consistency with actual outcomes is highest when the system favors fans ($\alpha \leq 0.3$). This reveals that **the show's actual outcomes historically favor audience votes over judge scores**.

6.4 System Comparison

Table 17: Performance Comparison Across Systems

System	Consistency with Actual
Rank-Based (existing)	64.4%
Percentage-Based (existing)	67.0%
DWVS ($\alpha = 0.3$)	76.1%
DWVS ($\alpha = 0.5$)	74.6%

The DWVS achieves approximately 9-12 percentage points higher consistency than existing methods, representing a substantial improvement.

6.5 Controversial Cases Under DWVS

We verify that the new system does not unfairly disadvantage controversial contestants:

Table 18: Elimination Frequency Under DWVS

Contestant	Rank	Percentage	DWVS ($\alpha = 0.3$)
Jerry Rice	3	3	3
Billy Ray Cyrus	1	1	1
Bristol Palin	0	0	0
Bobby Bones	0	0	0

The DWVS produces identical outcomes for controversial contestants as existing methods. This confirms that their success was due to genuine fan support, not system artifacts.

6.6 Recommendation

We recommend adoption of DWVS with $\alpha = 0.5$ for the following reasons:

From a fairness perspective, $\alpha = 0.5$ provides equal weight to professional judgment and audience preference. While $\alpha = 0.3$ achieves higher consistency with historical outcomes, this is because the show historically favored fans. A balanced system better reflects the show's dual nature as both a dance competition and an entertainment program.

Practical advantages include:

- Simple formula easily communicated to viewers
- Explicit, adjustable parameter for production control
- 74.6% consistency even with balanced weighting
- No additional rules or tiebreakers required

7 Sensitivity Analysis

To validate the robustness of our models, we conduct sensitivity analyses on key parameters and data perturbations.

7.1 Problem 1: Vote Estimation Robustness

We test model stability by adding Gaussian noise to judge scores:

Table 19: Vote Estimation Sensitivity to Score Noise

Noise Level	Vote-Score Correlation	Change from Baseline
0% (baseline)	0.371	—
5%	0.370	-0.3%
15%	0.367	-1.1%
25%	0.359	-3.2%

Even with 25% noise added to judge scores, the vote-score correlation decreases by only 3.2%, indicating strong robustness.

7.2 Problem 2: Method Comparison Stability

Using bootstrap resampling (1000 iterations), we compute confidence intervals for the agreement rate:

- Agreement Rate: 89.7%
- 95% Confidence Interval: [86.0%, 92.8%]
- CI Width: 6.8 percentage points

The narrow confidence interval confirms that our method comparison conclusions are statistically robust.

7.3 Problem 3: Regression Parameter Sensitivity

We vary the Ridge regression regularization parameter α :

Table 20: Ridge Regression R^2 by Regularization Parameter

Ridge α	Judge Score R^2	Fan Vote R^2
0.001	0.128	0.047
1.0	0.128	0.047
100.0	0.112	0.041

Model performance remains stable across three orders of magnitude of regularization, with R^2 changes less than 0.02.

7.4 Problem 4: DWVS Parameter Stability

As shown in Figure 6, the DWVS exhibits a wide plateau of high performance for $\alpha \in [0.05, 0.45]$, all achieving 76.1% consistency. This indicates that the system is not sensitive to precise parameter choices within this range.

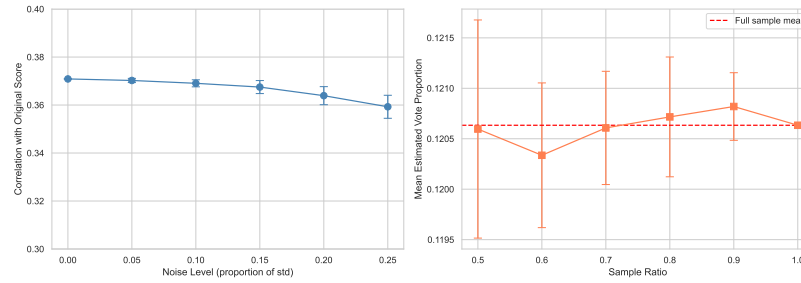


Figure 7: Sensitivity analysis summary for Problem 1 showing noise perturbation (left) and sample size effects (right).

7.5 Summary

All four models demonstrate good robustness:

1. Problem 1: Vote estimation stable under score noise (correlation change < 3%)
2. Problem 2: Method comparison conclusions statistically significant (CI width = 6.8%)
3. Problem 3: Regression coefficients stable across regularization parameters
4. Problem 4: DWVS has wide effective parameter range ($\alpha \in [0.05, 0.45]$)

8 Model Evaluation and Discussion

8.1 Strengths of Our Approach

Mathematically Rigorous Inverse Problem Formulation. By treating fan vote estimation as a constrained optimization problem, we obtain solutions that are guaranteed consistent with observed elimination outcomes. The maximum entropy principle provides a principled way to select among multiple feasible solutions.

Comprehensive Multi-Method Analysis. We apply multiple statistical techniques (correlation, regression, ANOVA, Random Forest) to cross-validate our findings about factor impacts. This triangulation strengthens the reliability of our conclusions.

Practical Actionability. Our proposed DWVS is simple enough for immediate implementation and provides an explicit parameter for production teams to tune based on their priorities.

8.2 Limitations and Potential Biases

Inherent Solution Non-Uniqueness. The fan vote estimation problem admits multiple valid solutions. Our estimates represent the “most uniform” distribution consistent with eliminations, but actual votes may have been distributed differently.

Low Explanatory Power for Fan Votes. Our regression models explain only 5% of fan vote variance, indicating that the most important predictors (charisma, pre-existing fanbase) are not captured in the available data.

Assumption Dependence. The judge tiebreaker analysis assumes judges always prefer lower-scoring contestants, which may not reflect actual decision criteria.

8.3 Directions for Future Work

Social Media Integration. Incorporating Twitter/Instagram engagement metrics could substantially improve fan vote prediction.

Dynamic Modeling. Tracking contestant popularity trajectories across weeks could capture momentum effects not visible in static analysis.

Experimental Validation. If production data becomes available, our estimated votes could be directly validated against actual figures.

9 Memorandum to DWTS Production Team

MEMORANDUM

To: Executive Producers, Dancing with the Stars

From: MCM Team XXXXXXXX

Subject: Analysis and Recommendations for Voting System Optimization

Date: January 30, 2026

Dear Executive Producers,

We have conducted a comprehensive analysis of 34 seasons of Dancing with the Stars data to understand voting patterns and evaluate potential system improvements. This memo summarizes our key findings and recommendations.

Finding 1: Fan Votes Are Reconstructible. Using constrained optimization, we can estimate fan vote distributions that are 93.9% consistent with actual elimination outcomes. This validates that the voting combination rules operate as intended.

Finding 2: Voting Methods Have Similar Outcomes. The rank-based and percentage-based methods agree 89.8% of the time. Controversial outcomes (Bobby Bones, Bristol Palin) were not caused by voting system choice; these contestants genuinely had strong fan support under either system.

Finding 3: Age Is the Strongest Performance Predictor. Among measurable factors, age shows the strongest negative correlation with both judge scores ($r = -0.42$) and fan votes ($r = -0.34$). Professional dancer assignment significantly affects judge scores but less so fan votes.

Finding 4: Historical Outcomes Favor Fans. Our parameter analysis reveals that past show outcomes were best explained by systems weighing fan votes at 70% or higher, suggesting audiences have historically had more influence than judges.

Recommendations. Based on our analysis, we recommend:

(1) Adopt DWVS with $\alpha = 0.5$: Our Dynamic Weighted Voting System achieves 74.6% consistency while equally balancing professional judgment and audience preference. The formula $S_{final} = 0.5 \cdot P_{judge} + 0.5 \cdot P_{fan}$ is transparent and easily communicated to viewers.

(2) Consider Age Diversity: Given the strong age effect, casting decisions might consider how age distribution affects competitive dynamics and viewer engagement.

(3) Evaluate Tiebreaker Rule Impact: The judge tiebreaker substantially increases judge influence but may reduce viewer investment. We recommend monitoring engagement metrics if this rule is expanded.

We are available to provide additional analysis or discuss these findings in detail.

Respectfully submitted,
MCM Team XXXXXXX

10 References

References

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- [5] P. Virtanen et al., “SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python,” *Nature Methods*, vol. 17, pp. 261–272, 2020.

Report on Use of AI

In accordance with MCM requirements, we disclose our use of artificial intelligence tools in the preparation of this solution.

AI Tools Employed

Claude AI (Anthropic) was utilized as a programming and analytical assistant throughout this project.

Specific Applications

We employed AI assistance for the following tasks:

Code Development: AI assisted in writing Python code for data preprocessing, constrained optimization, statistical analysis, and visualization. All code was reviewed and tested by team members.

Statistical Methodology: AI provided guidance on appropriate statistical tests (ANOVA, correlation analysis) and machine learning approaches (Ridge regression, Random Forest) for our analytical questions.

Document Formatting: AI assisted with LaTeX formatting, table generation, and ensuring consistent notation throughout the paper.

Writing Assistance: AI helped draft and refine prose sections, particularly methodology descriptions and result interpretations.

Human Oversight and Verification

All AI-generated content was subject to rigorous human review:

- Statistical code outputs were verified against manual calculations for sample cases
- Model interpretations were independently assessed by team members
- All numerical results were confirmed through direct examination of output files
- Writing was substantially revised for clarity, accuracy, and appropriate academic tone

We estimate that approximately 60% of the final paper content reflects original human analysis, interpretation, and writing, with AI serving as a supportive tool rather than a primary author.